WORKED SOLUTION

STEP III 2012, Question 7

Pure • Coupled Differential Equations

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s3-q7

The key idea

Two coupled first-order equations become one second-order equation once we eliminate a variable: differentiate the $\dot{y}$ equation, then use the first equation itself to write z in terms of y and $\dot{y}$. After that, each part is just choosing the constants A and B. In part (iii) the infinite sum is two geometric series, and matching the result with part (ii) pins down c. Physically, z_2 is the steady state reached by giving the same injection once every unit of time.

Solution

The general solution

Differentiating $\dot{y} = -2y + 2z$ and then substituting the second equation for $\dot{z}$,

$$\ddot{y} = -2\dot{y} + 2\dot{z} = -2\dot{y} + 2(-\dot{y} - 3z) = -4\dot{y} - 6z.$$

The first equation gives z in terms of y:

$$z = y + \tfrac{1}{2}\dot{y}, \qquad \text{so} \qquad 6z = 6y + 3\dot{y}.$$

Substituting,

$$\ddot{y} + 7\dot{y} + 6y = 0.$$

The auxiliary equation $m^2 + 7m + 6 = (m+1)(m+6) = 0$ has roots $m = -1$ and $m = -6$, so

$$y = Ae^{-t} + Be^{-6t}$$

for arbitrary constants A and B. Then z follows from $z = y + \frac{1}{2}\dot{y}$:

$$z = Ae^{-t}(1 - \tfrac{1}{2}) + Be^{-6t}(1 - 3) = \tfrac{1}{2}Ae^{-t} - 2Be^{-6t},$$

as required.

Ask yourself We used both equations to get the equation for y, but only the first to find z. Does the pair we found really satisfy the second equation?

It does. With $z = y + \frac{1}{2}\dot{y}$ we have $\dot{z} = \dot{y} + \frac{1}{2}\ddot{y} = \dot{y} + \frac{1}{2}(-7\dot{y} - 6y) = -\frac{5}{2}\dot{y} - 3y$. Also $-\dot{y} - 3z = -\dot{y} - 3y - \frac{3}{2}\dot{y} = -\frac{5}{2}\dot{y} - 3y$, which is the same. So every solution of the system has this form, and every pair of this form is a solution.

Part (i)

At $t = 0$ we have $y(0) = A + B$ and $z(0) = \frac{1}{2}A - 2B$. The conditions give

$$\tfrac{1}{2}A - 2B = 0, \qquad A + B = 5,$$

so $A = 4B$, and then $5B = 5$. Hence $B = 1$ and $A = 4$:

$$y = 4\mathrm{e}^{-t} + \mathrm{e}^{-6t}, \qquad z_1(t) = 2\mathrm{e}^{-t} - 2\mathrm{e}^{-6t}.$$

The drug starts in the blood, the amount in the brain rises from zero, and then it decays as it is absorbed.

Part (ii)

Write $a = \frac{1}{2}A$ and $b = 2B$, so that $z = a\mathrm{e}^{-t} - b\mathrm{e}^{-6t}$. The conditions are

$$a - b = c, \qquad a\mathrm{e}^{-1} - b\mathrm{e}^{-6} = c.$$

Subtracting the second from the first gives $a(1 - \mathrm{e}^{-1}) = b(1 - \mathrm{e}^{-6})$. Call this common value k, so that

$$a = \frac{k}{1 - \mathrm{e}^{-1}}, \qquad b = \frac{k}{1 - \mathrm{e}^{-6}}, \qquad \text{and} \qquad c = a - b = kC,$$

where

$$C = \frac{1}{1 - \mathrm{e}^{-1}} - \frac{1}{1 - \mathrm{e}^{-6}}.$$

Since $\mathrm{e}^{-1} > \mathrm{e}^{-6}$, we have $1 - \mathrm{e}^{-1} < 1 - \mathrm{e}^{-6}$, so $C > 0$. In particular $C \neq 0$, and we may divide by it: $k = c/C$. (Equivalently, the two linear equations have determinant $\mathrm{e}^{-1} - \mathrm{e}^{-6} \neq 0$, so the solution is unique.) Therefore

$$z_2(t) = \frac{c}{C}\left(\frac{\mathrm{e}^{-t}}{1 - \mathrm{e}^{-1}} - \frac{\mathrm{e}^{-6t}}{1 - \mathrm{e}^{-6}}\right),$$
$$y_2(t) = \frac{c}{C}\left(\frac{2\mathrm{e}^{-t}}{1 - \mathrm{e}^{-1}} + \frac{\mathrm{e}^{-6t}}{2(1 - \mathrm{e}^{-6})}\right),$$

where y_2 comes from $A = 2a$ and $B = \frac{1}{2}b$.

Part (iii)

Put $m = -n$, so that m runs over $0, 1, 2, \ldots$ and $t - n = t + m$. Using part (i),

$$\sum_{n=-\infty}^{0} z_1(t-n) = \sum_{m=0}^{\infty} (2\mathrm{e}^{-t}(\mathrm{e}^{-1})^m - 2\mathrm{e}^{-6t}(\mathrm{e}^{-6})^m).$$

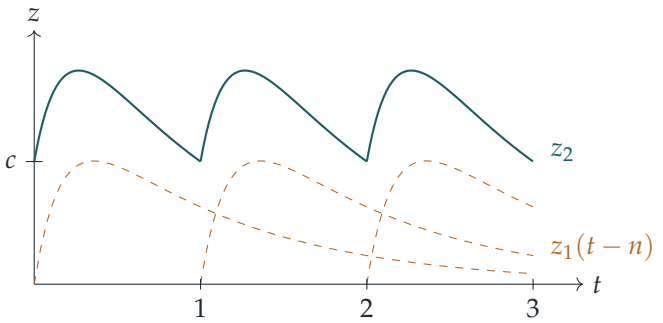
Both pieces are geometric series with ratios e^{-1} and e^{-6}, which lie strictly between 0 and 1, so each converges and we may sum them separately:

$$\sum_{n=-\infty}^{0} z_1(t-n) = \frac{2\mathrm{e}^{-t}}{1-\mathrm{e}^{-1}} - \frac{2\mathrm{e}^{-6t}}{1-\mathrm{e}^{-6}}.$$

This is exactly the bracket in $z_2(t)$, multiplied by 2. So the identity holds for $0 \le t \le 1$ precisely when $c/C = 2$, that is, when

$$c = 2C = \frac{2}{1-\mathrm{e}^{-1}} - \frac{2}{1-\mathrm{e}^{-6}} \approx 1.159.$$

(For any other c the two sides differ by a non-zero multiple of the same non-zero function, so this value is the only one.)



The picture shows the meaning. A dose of 5 injected into the blood at time n produces the brain curve $z_1(t-n)$ (dashed). If a dose is given at every integer time, from the distant past, the brain quantity is the sum of all these responses. That sum repeats every unit of time (solid curve), and on $[0, 1]$ it is z_2 with this value of c.

Checks

- In part (i), $z_1(0) = 2 - 2 = 0$ and $y(0) = 4 + 1 = 5$.
- In part (iii), $z_2(0)$ and $z_2(1)$ should both equal c. In the sum, the $t = 1$ series is the $t = 0$ series with its first term removed, and that term is $z_1(0) = 0$. This confirms $z(0) = z(1)$ without any algebra.
- With $c = 2C$, $y_2(0) - y_2(1) = 4 + 1 = 5$: over each period the blood loses exactly one dose, which the next injection replaces.
- Numerically, $c \approx 1.15898$. Summing 200 terms of $z_1(t+m)$ at $t = 0.37$ gives $1.967693\ldots$, matching $z_2(0.37)$ to 12 decimal places.

Common slips

- Forgetting that z is fixed by y through $z = y + \frac{1}{2}\dot{y}$, and solving a second equation for z with new, independent constants.
- Getting the direction of the shift wrong in part (iii): $n \leq 0$ means $t - n \geq t$, so only doses from the past contribute.

Beyond STEP

This is a linear compartment model, written as $\dot{\mathbf{x}} = M\mathbf{x}$ with a 2×2 matrix M. The decay rates 1 and 6 are minus the eigenvalues of M, which is the matrix-exponential view of linear systems that you meet in a first-year course. Part (iii) is a simple case of superposition: the response to repeated impulses is a sum of shifted copies of one impulse response. This idea leads to Green's functions and convolution. Boyce and DiPrima, *Elementary Differential Equations*, covers systems of linear equations and impulse responses.

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